2 Background and Related Work

New training data affect more than a continual learner's performance on its current task. They can also change what it remembers from earlier tasks. This makes the training data a possible route for an attack on knowledge the model has already acquired. Detecting such an attack requires some understanding of how the learner updates, what the attacker can change, and which model responses are available before training begins.

2.1 Continual Learning and Forgetting

Continual learning studies models that learn from a sequence of tasks or data distributions. The difficulty is to learn new information while retaining earlier knowledge. Training on a new task can change shared parameters in ways that substantially reduce accuracy on previous tasks, an effect known as catastrophic forgetting [1]. Some protocols allow earlier examples to be stored and replayed; others require learning to continue without them.

Let $D_t = \{(x_i, y_i)\}_{i=1}^{N_t}$ denote the training set of task t, with tasks indexed from zero. A learner enters task t with parameters θ_{t-1} and produces θ_t after training on D_t. Minimising the current empirical loss,

$$\mathcal{L}_t(\theta) = \frac{1}{N_t} \sum_{(x,y)\in D_t} \ell\big(f_\theta(x, t), y\big), \tag{1}$$

places no explicit constraint on performance on $D_0, \ldots, D_{t-1}$. Here the argument t allows a task-specific output head to be selected; the class of x remains to be predicted.

The balance between retaining earlier knowledge and adapting to new data is usually described as the stability–plasticity trade-off. Replay methods revisit stored or generated examples. Parameter-isolation methods allocate model capacity to different tasks. Regularisation methods penalise changes to parameters or predictions associated with earlier knowledge [1]. The experiments here focus on this last group and examine its behaviour when the new training data are deliberately modified.

2.1.1 Task Identity and Class Labels

Van de Ven and Tolias distinguish three evaluation scenarios according to the information available at test time [2]. In task-incremental learning, task identity is provided and can select the appropriate output head. In class-incremental learning, the model must classify among the classes encountered so far without being told the task. In domain-incremental learning, the input distribution changes while the required output space remains shared, and task identity is not provided. These scenarios impose different demands on the classifier and need to be distinguished when comparing results.

The learner used here is task-incremental and has a separate head for each task. The incoming task and its output space are known, but the detector must score images without their true class labels. Its supervised training uses a different label: whether an image is clean or artificially poisoned. Those labels are available in the controlled source experiment and absent when a new dataset is scored. Task identity, image class and clean/poison status therefore have separate roles throughout the thesis.

2.1.2 Measuring Retention

Let $a_{t,j}$ be the test accuracy on task j after learning task t. The resulting accuracy matrix records both learning and retention. Following the backward-transfer formulation of Lopez-Paz and Ranzato [3], the average change on earlier tasks after learning task $t \geq 1$ is

$$\mathrm{BWT}_t = \frac{1}{t} \sum_{j=0}^{t-1} (a_{t,j} - a_{j,j}). \tag{2}$$

A negative value means that earlier-task accuracies have declined, on average, from their immediate post-training values. The mean historical accuracy is

$$A_t^{\mathrm{past}} = \frac{1}{t} \sum_{j=0}^{t-1} a_{t,j}, \tag{3}$$

which measures the performance that remains after the update. Both quantities are useful: a small backward-transfer loss can hide poor historical accuracy if the earlier tasks were never learned well.

Since clean sequential training can itself cause forgetting, poisoning needs a paired control. Starting from the same pre-update checkpoint, the learner is trained separately on clean and poisoned versions of the incoming task. The difference $A_{t,\mathrm{clean}}^{\mathrm{past}} - A_{t,\mathrm{poison}}^{\mathrm{past}}$ then describes the extra historical accuracy loss attributable to the changed data in that experiment. Measuring the benefit of a detector would require a further training run in which its decisions are used to filter the data or modify learning.

2.2 Regularisation-Based Continual Learning

Several methods considered here use a parameter penalty of the general form

$$\mathcal{L}_t^{\mathrm{CL}}(\theta) = \mathcal{L}_t(\theta) + \frac{\lambda}{2} \sum_r \Omega_{t-1,r} (\theta_r - \bar{\theta}_{t-1,r})^2. \tag{4}$$

Here r indexes scalar parameters, $\bar{\theta}_{t-1}$ is a retained reference parameter vector, and $\Omega_{t-1,r}$ measures importance. Methods differ in how these quantities are estimated and updated. A larger penalty restricts movement in important coordinates, preserving earlier knowledge at the possible cost of slower adaptation to the new task.

2.2.1 Elastic Weight Consolidation

Elastic Weight Consolidation (EWC) estimates parameter importance using a diagonal approximation to the Fisher information [4]. For a previous task A, its two-task penalty is

$$\mathcal{L}_{\mathrm{EWC}}(\theta) = \mathcal{L}_t(\theta) + \frac{\lambda}{2} \sum_r F_{A,r} (\theta_r - \theta_{A,r}^*)^2. \tag{5}$$

Larger Fisher values give stronger constraints. The penalty follows from a local approximation to the posterior after the earlier task. Using only the diagonal makes it practical to compute, at the cost of omitting interactions between parameters. Extending EWC to a task sequence also requires choosing how to retain or combine importance estimates.

EWC's accumulated importance values control the training update. By contrast, the detector's gradient norms describe the response of a frozen model to one image. Although both involve gradients, they measure different quantities.

2.2.2 Memory Aware Synapses and Riemannian Walk

Memory Aware Synapses (MAS) measures importance through the sensitivity of the network's output function to parameter changes [5]. Unlike a loss-based importance estimate, this calculation can use unlabelled inputs. Parameters judged important are then protected during subsequent learning. The unlabelled part of MAS is this importance estimate; the task classifier is still trained with class labels.

Riemannian Walk (RWalk) combines Fisher-based information with a training-path contribution to parameter importance [6]. Its formulation connects changes in model distributions with the optimisation trajectory. Chaudhry et al. also distinguish forgetting from intransigence, the difficulty of acquiring new knowledge. A method that barely changes its parameters may retain earlier predictions but learn the incoming task poorly, so retention and new-task accuracy need to be examined together.

2.2.3 Auxiliary-Network Approaches

Active Forgetting with synaptic Expansion–Convergence (AFEC) addresses negative transfer from earlier knowledge to a new task [7]. It introduces additional parameters for learning the new task and then selectively combines them with the existing network. This allows earlier constraints to be relaxed where they obstruct learning the new task.

Auxiliary Network Continual Learning (ANCL) also introduces an auxiliary network, but uses it to provide a plasticity-oriented reference for the main learner [8]. Its regulariser balances the previous model and the auxiliary solution. The shared use of an auxiliary network thus serves different formulations in AFEC and ANCL. Their EWC-based variants are included in the experimental comparison.

2.3 Data Poisoning and Forgetting-Inducing Attacks

In a poisoning attack, the adversary changes data on which the victim will train. In an evasion attack, the adversary instead changes an input to a fixed, trained model. Poisoning therefore acts through the learner's update: the modified data must cause damage after they have been used for training. For example, Biggio et al. optimise an injected SVM training point to increase a downstream validation loss [9]. An input's difficulty before training is only part of this problem.

A poisoning objective can be written schematically as

$$\max_{\Delta \in \mathcal{C}} \quad \mathcal{J}\big(\theta^{+}(\Delta)\big), \tag{6}$$

$$\text{where} \quad \theta^{+}(\Delta) = \mathcal{A}(\theta^{-}, D_t^{\Delta}). \tag{7}$$

Here $\mathcal{A}$ is the victim's update procedure, θ^{-} is its pre-update state, and $\mathcal{C}$ specifies the attacker's permitted changes. The outer objective $\mathcal{J}$ measures the attacker's desired damage. The update operator $\mathcal{A}$ can represent a finite training schedule, as well as an approximate solution to a learning objective.

For additive image poisoning, the modified input can be represented as $\widetilde{x}_i = \Pi_{[0,1]^d}(x_i + \delta_i)$, with $\|\delta_i\|_\infty \leq \varepsilon$. The projection keeps pixel values in range. The budget ε bounds the change to each pixel. The poisoning fraction specifies how many images are changed. These are separate settings: small perturbations may be applied to every image, or larger perturbations to only a few.

2.3.1 BrainWash

BrainWash targets the retention of earlier tasks rather than only the accuracy of the incoming task [10]. The attacker has access to the learned model and incoming-task data, but not the original historical training sets. It first synthesises historical inputs through model inversion, then uses them to optimise perturbations that make a subsequent update damage earlier-task performance.

In the reckless setting, the attacker seeks forgetting without requiring good performance on the incoming task. In the cautious setting, the attack also seeks to preserve incoming-task accuracy, so monitoring that accuracy alone may miss the damage. Both use constrained input perturbations, optimised through an approximation to a bilevel objective [10].

For detection, the relevant point is that damage to historical tasks may remain hidden by acceptable performance on the new task. Historical retention and incoming-task accuracy are therefore measured separately. The attack budgets and optimisation settings used here are given in the experimental setup.

2.3.2 Poisoning Detection and Existing Defences

Backdoor defences provide related but different detection settings. Spectral Signatures identifies unusual directions in learned representations and uses them to detect and remove poisoned examples [11]. This work uses representations to reveal contamination that ordinary test accuracy can overlook. Its setting is a classifier trained on backdoored data, whereas the detector here examines a new task before learning it. Whether comparable signals exist at this earlier stage must be tested.

A detection score also needs to be distinguished from a training defence. The former identifies suspicious data; the latter acts on the learning procedure, for example by filtering images or constraining updates. The experiments build on the PROACT repository, which already contains training-time defence components [12]. The detector is a separate addition. Assessing whether its alerts improve retention would require using them during training, beyond measuring detection and attack effectiveness separately.

2.4 Model Inversion as Historical Reference Data

Model inversion synthesises inputs using information contained in a trained model. DeepInversion optimises image inputs while keeping the network fixed [13]. A classification objective encourages the requested class, while image priors and BatchNorm feature-statistic constraints guide synthesis. The stored BatchNorm statistics help constrain the otherwise underdetermined search: many inputs can produce a desired network output.

The resulting images are synthetic inputs, rather than recovered copies of the original training set. Matching selected feature statistics still leaves room for differences from real images. Smith et al. examine this difficulty in data-free class-incremental learning [14]: replaying inverted images with standard distillation is insufficient on common benchmarks, motivating changes to the learning procedure. Their problem is different from detection, but it illustrates why the quality of a synthetic reference needs to be examined rather than assumed.

Inversion serves two purposes in these experiments. It supplies the attacker with a surrogate for inaccessible historical data, and supplies the label-free detector with historical reference images. These uses place different demands on the samples. Images that help

construct a damaging update need not be representative enough to calibrate a clean-data test. The reference-based detector consequently relies both on the historical checkpoints being clean and on the adequacy of the synthetic references.

2.5 Model-Based Descriptors and Detection Scores

The frozen learner provides several ways to describe an incoming image. Its output probabilities give the head's response, activations give its internal representation, and loss gradients measure local parameter sensitivity. These descriptors can be supplied to a classifier or compared across datasets without training the learner on the incoming images.

2.5.1 Gradients and Their Limits

For a differentiable loss, write $g(x, y) = \nabla_\theta \ell(f_\theta(x), y)$. A small gradient step on (x, y) changes parameters approximately by $-\eta g(x, y)$. If $\mathcal{L}_{\text{past}}$ is a differentiable historical loss, a first-order expansion gives

$$\mathcal{L}_{\text{past}}(\theta - \eta g) - \mathcal{L}_{\text{past}}(\theta) \approx -\eta \, \nabla_\theta \mathcal{L}_{\text{past}}(\theta)^\top g. \tag{8}$$

The inner product determines whether this small step increases or decreases historical loss. Gradient magnitude alone omits that direction information. This is a local explanation of interference; a full training update also involves successive steps, regularisation and changes in the representation.

Influence functions provide a related approach to studying how training examples affect model predictions [15]. They account for local curvature through the training objective, rather than assigning influence from a raw gradient norm alone. The descriptors used here retain less information: they summarise gradients without estimating the curvature or the eventual effect of an image on training.

Gradient magnitude can be summarised at different scales. If the backbone parameters are partitioned into tensors $\theta^{(1)}, \ldots, \theta^{(q)}$, one can record $\|\nabla_{\theta^{(j)}} \ell\|_2$ for every tensor. This is one feature for an entire weight or bias tensor, not one feature per scalar parameter. A stage norm instead combines several tensors. For a disjoint partition,

$$\|\nabla_\theta \ell\|_2^2 = \sum_{j=1}^{q} \|\nabla_{\theta^{(j)}} \ell\|_2^2. \tag{9}$$

Tensor norms retain the distribution of gradient magnitude within a stage, which the stage norm combines into a single value. They also increase dimensionality and depend on tensor size and parameterisation. Comparing the two representations tests whether that additional detail helps detection.

Computing a loss gradient normally requires a class target. Since the detector lacks the true incoming class, it uses the model's predicted class as a fixed pseudo-target for differentiation. The gradient then measures sensitivity relative to that prediction,

including when it is incorrect. The method chapter specifies this construction and the fixed incoming head used to obtain it.

2.5.2 Predictions and Activations

For predicted probabilities p, entropy is $H(p) = -\sum_c p_c \log p_c$. Other summaries are the maximum probability and the gap between the two largest values. Hendrycks and Gimpel use maximum softmax probability as a baseline for identifying misclassified and out-of-distribution inputs [16]. Low confidence can thus be useful for identifying unfamiliar inputs, although unfamiliarity can arise from clean task changes as well as poisoning.

Guo et al. [17] show that modern trained neural networks can produce probabilities that are poorly calibrated to classification correctness. Here the incoming head has not yet been trained on its task, so its probability vector is used as a model response rather than a calibrated estimate of correctness. The activation norm adds a measurement of representation magnitude. Any association between these responses and poisoning is learned or tested in the detection experiment.

2.5.3 Ranking, Calibration and Dataset Decisions

A supervised detector maps descriptors $z(x)$ to a scalar score, for example through logistic regression,

$$s(x) = \sigma\big(w^\top \widetilde{z}(x) + b\big), \qquad \sigma(u) = \frac{1}{1 + e^{-u}}, \tag{10}$$

where $\widetilde{z}$ denotes descriptors transformed using source-fitted scaling. The binary target distinguishes clean and poisoned images. The score lies between zero and one, but its probabilistic interpretation depends on the training population and class proportions. A balanced source experiment alone cannot calibrate it for deployment on a different task.

ROC analysis evaluates how examples are ordered as the decision threshold varies [18]. A high ROC-AUC can therefore coexist with poor decisions at one frozen cutoff. After task transfer, for example, clean and poisoned scores may both shift while their relative ordering remains largely unchanged.

Probability calibration concerns agreement between predicted probabilities and observed frequencies [17]. Threshold calibration has a narrower purpose: choosing a cutoff for a decision rule. Here that cutoff is estimated from historical data and frozen before the target task. Its error rate on the target remains something to measure.

A sample score can be turned into a dataset alert by counting the images above a cutoff u:

$$K(D) = \sum_{i=1}^{n} \mathbf{1}\{s(x_i) \geq u\}. \tag{11}$$

A count rule alerts when $K(D) \geq k$. Under independent clean-image alarms with a common probability p, the count has a Binomial(n, p) distribution. A one-sided Clopper–Pearson bound can account for uncertainty in p estimated from a separate clean sample [19]. This calculation relies on the clean-image alarm probability remaining stable. Task shift can violate that assumption. The number of distinct calibration images also matters: repeatedly drawing datasets from the same finite pool does not supply new independent images.

2.6 Historical-Reference Distribution Tests

Without labelled clean/poison examples, historical data offer a possible reference for incoming descriptors. Such a comparison tests a property of the data distributions, rather than the cause of a difference. New classes and imperfect inversion samples can produce clean distribution shifts; conversely, poisoning may leave the property being tested almost unchanged.

2.6.1 Maximum Mean Discrepancy

Maximum mean discrepancy (MMD) compares distributions through their embeddings in a reproducing kernel Hilbert space [20]. For independent $Z, Z' \sim P$ and $W, W' \sim Q$, its squared population value is

$$\mathrm{MMD}^2(P, Q) = \mathbb{E}k(Z, Z') + \mathbb{E}k(W, W') - 2\mathbb{E}k(Z, W). \tag{12}$$

A characteristic kernel distinguishes unequal distributions at the population level. A finite-sample test still depends on sample size, kernel choice and the reference data. Random Fourier features approximate shift-invariant kernels by finite-dimensional inner products, reducing the cost of repeated comparisons [21].

This sensitivity also presents a difficulty for historical references. Synthetic historical images and real incoming images may differ even when both are clean. In that case, rejecting equality is consistent with the statistical question but unhelpful for identifying poisoning. A permutation test requires exchangeability under its null, which the clean status of the two groups alone cannot establish. Clean-alert and attack-detection rates are therefore both needed to assess this baseline.

2.6.2 Kendall Correlations and Rank Dependence

Kendall's rank correlation compares the ordering of paired observations [22]. For descriptor coordinates a and b, the tau-a statistic is

$$\widehat{\tau}_{ab} = \frac{2}{n(n-1)} \sum_{i<j} \operatorname{sign}(z_{ia} - z_{ja}) \operatorname{sign}(z_{ib} - z_{jb}). \tag{13}$$

Concordant orderings contribute $+1$, discordant orderings -1, and ties zero. Tau-a is not the tie-normalised tau-b statistic. A vector of these coefficients describes pairwise rank dependence across a dataset.

Strictly increasing transformations of individual coordinates leave these orderings unchanged. A rank comparison ignores these marginal changes, allowing some scale differences between tasks to be removed. It also misses an attack expressed entirely through such transformations. More generally, pairwise Kendall coefficients describe only part of the joint distribution; higher-order dependence may change while these coefficients stay fixed.

I use the name *Rank* for the historical-reference test built from these coefficients. The descriptors are measured per image, but the correlations summarise a whole dataset. This gives a dataset-level comparison without first classifying individual images as clean or poisoned. The test statistic and the rule for combining historical comparisons are described in the method chapter.

2.6.3 Jackknife Variation and Bootstrap Approximation

Correlation estimates vary with the sampled images. The jackknife measures this variation by leaving out observations in turn [23]. For a statistic T_n and its leave-one-out value $T_{n,-i}$, the pseudovalue is

$$P_i = nT_n - (n-1)T_{n,-i}. \tag{14}$$

Variation across the pseudovalues estimates first-order variability in the statistic. When that variation is effectively zero, a first-order approximation becomes degenerate and needs separate treatment.

Gaussian multiplier methods approximate sampling errors using random weights applied to estimated observation contributions. Chen develops Gaussian and bootstrap approximations for nondegenerate, high-dimensional U-statistics, including jackknife-based multiplier procedures [24]. Kendall coefficients are order-two U-statistics, which makes this framework relevant to comparing correlation profiles. Applying it here still requires assumptions about independence and nondegeneracy. These are particularly important for synthetic inversion samples. The implemented two-sample procedure is an approximation, rather than a finite-sample guarantee supplied by Chen's result.

The bootstrap draws simulate fluctuations of the statistic conditional on the observed images; the image sample itself stays fixed. The resulting p-value concerns a null hypothesis about rank dependence, not the probability of poisoning. Rejection and non-rejection must be interpreted within that scope, with non-rejection leaving open changes that the test has little power to detect.

2.7 Scope of This Thesis

The remaining question is how useful these measurements are before a new task is learned. The supervised approach asks whether source-trained scores and historical thresholds transfer to a later task. The label-free approach asks whether inversion references support useful dataset comparisons without examples labelled as poisoned. Both depend on how model responses change across tasks, but they use different information and make different assumptions.

The CL algorithms, BrainWash attack and inversion procedure are existing components. This work develops the detector around them, compares gradient granularities and feature groups, and examines where the resulting decisions work or fail. The next chapter defines the information available to the detector and attacker before describing these methods in detail.

References

[1] German I. Parisi, Ronald Kemker, Jose L. Part, Christopher Kanan, and Stefan Wermter. Continual lifelong learning with neural networks: A review. *Neural Networks*, 113:54–71, 2019. doi: 10.1016/j.neunet.2019.01.012.

[2] Gido M. van de Ven and Andreas S. Tolias. Three scenarios for continual learning. arXiv:1904.07734, 2019. URL https://arxiv.org/abs/1904.07734.

[3] David Lopez-Paz and Marc'Aurelio Ranzato. Gradient episodic memory for continual learning. In *Advances in Neural Information Processing Systems*, volume 30, 2017. URL https://proceedings.neurips.cc/paper_files/paper/2017/hash/f87522788a2be2d171666752f97ddebb-Abstract.html.

[4] James Kirkpatrick, Razvan Pascanu, Neil Rabinowitz, Joel Veness, Guillaume Desjardins, Andrei A. Rusu, Kieran Milan, John Quan, Tiago Ramalho, Agnieszka Grabska-Barwinska, Demis Hassabis, Claudia Clopath, Dharshan Kumaran, and Raia Hadsell. Overcoming catastrophic forgetting in neural networks. *Proceedings of the National Academy of Sciences*, 114(13):3521–3526, 2017. doi: 10.1073/pnas.1611835114.

[5] Rahaf Aljundi, Francesca Babiloni, Mohamed Elhoseiny, Marcus Rohrbach, and Tinne Tuytelaars. Memory aware synapses: Learning what (not) to forget. In *Proceedings of the European Conference on Computer Vision*, 2018. URL https://openaccess.thecvf.com/content_ECCV_2018/html/Rahaf_Aljundi_Memory_Aware_Synapses_ECCV_2018_paper.html.

[6] Arslan Chaudhry, Puneet K. Dokania, Thalaiyasingam Ajanthan, and Philip H. S. Torr. Riemannian walk for incremental learning: Understanding forgetting and intransigence. In *Proceedings of the European Conference on Computer Vision*, 2018. doi: 10.1007/978-3-030-01252-6_33.

[7] Liyuan Wang, Mingtian Zhang, Zhongfan Jia, Qian Li, Chenglong Bao, Kaisheng Ma, Jun Zhu, and Yi Zhong. AFEC: Active forgetting of negative transfer in continual learning. In *Advances in Neural Information Processing Systems*, volume 34, 2021. URL https://proceedings.neurips.cc/paper/2021/hash/bc6dc48b743dc5d013b1abaebd2faed2-Abstract.html.

[8] Sanghwan Kim, Lorenzo Noci, Antonio Orvieto, and Thomas Hofmann. Achieving a better stability-plasticity trade-off via auxiliary networks in continual learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023. URL https://arxiv.org/abs/2303.09483.

[9] Battista Biggio, Blaine Nelson, and Pavel Laskov. Poisoning attacks against support vector machines. In *Proceedings of the 29th International Conference on Machine Learning*, 2012. URL https://arxiv.org/abs/1206.6389.

[10] Ali Abbasi, Parsa Nooralinejad, Hamed Pirsiavash, and Soheil Kolouri. BrainWash: A poisoning attack to forget in continual learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 24057–24067, 2024. URL https://openaccess.thecvf.com/content/CVPR2024/html/Abbasi_BrainWash_A_Poisoning_Attack_to_Forget_in_Continual_Learning_CVPR_2024_paper.html.

[11] Brandon Tran, Jerry Li, and Aleksander Madry. Spectral signatures in backdoor attacks. In *Advances in Neural Information Processing Systems*, volume 31, 2018. URL https://proceedings.neurips.cc/paper_files/paper/2018/hash/280cf18baf4311c92aa5a042336587d3-Abstract.html.

[12] PROACT. PROACT: Projection and activation constrained training for poisoning-resilient continual learning. Software repository, n.d. URL https://github.com/Choupoo/PROACT. Repository used for the experimental implementation; accessed 20 October 2026.

[13] Hongxu Yin, Pavlo Molchanov, Jose M. Alvarez, Zhizhong Li, Arun Mallya, Derek Hoiem, Niraj K. Jha, and Jan Kautz. Dreaming to distill: Data-free knowledge transfer via DeepInversion. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 8715–8724, 2020. URL https://openaccess.thecvf.com/content_CVPR_2020/html/Yin_Dreaming_to_Distill_Data-Free_Knowledge_Transfer_via_DeepInversion_CVPR_2020_paper.html.

[14] James Smith, Yen-Chang Hsu, Jonathan Balloch, Yilin Shen, Hongxia Jin, and Zsolt Kira. Always be dreaming: A new approach for data-free class-incremental learning. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 9374–9384, 2021. URL https://openaccess.thecvf.com/content/ICCV2021/html/Smith_Always_Be_Dreaming_A_New_Approach_for_Data-Free_Class-Incremental_Learning_ICCV_2021_paper.html.

[15] Pang Wei Koh and Percy Liang. Understanding black-box predictions via influence functions. In *Proceedings of the 34th International Conference on Machine Learning*, volume 70 of *Proceedings of Machine Learning Research*, pages 1885–1894, 2017. URL https://proceedings.mlr.press/v70/koh17a.html.

[16] Dan Hendrycks and Kevin Gimpel. A baseline for detecting misclassified and out-of-distribution examples in neural networks. In *International Conference on Learning Representations*, 2017. URL https://arxiv.org/abs/1610.02136.

[17] Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On calibration of modern neural networks. In *Proceedings of the 34th International Conference on Machine Learning*, volume 70 of *Proceedings of Machine Learning Research*, pages 1321–1330, 2017. URL https://proceedings.mlr.press/v70/guo17a.html.

[18] Tom Fawcett. An introduction to ROC analysis. *Pattern Recognition Letters*, 27(8):861–874, 2006. doi: 10.1016/j.patrec.2005.10.010.

[19] C. J. Clopper and E. S. Pearson. The use of confidence or fiducial limits illustrated in the case of the binomial. *Biometrika*, 26(4):404–413, 1934. doi: 10.1093/biomet/26.4.404.

[20] Arthur Gretton, Karsten M. Borgwardt, Malte J. Rasch, Bernhard Schölkopf, and Alexander Smola. A kernel two-sample test. *Journal of Machine Learning Research*, 13:723–773, 2012. URL https://www.jmlr.org/papers/v13/gretton12a.html.

[21] Ali Rahimi and Benjamin Recht. Random features for large-scale kernel machines. In *Advances in Neural Information Processing Systems*, volume 20, 2007. URL https://proceedings.neurips.cc/paper_files/paper/2007/hash/013a006f03dbc5392effeb8f18fda755-Abstract.html.

[22] M. G. Kendall. A new measure of rank correlation. *Biometrika*, 30(1–2):81–93, 1938. doi: 10.1093/biomet/30.1-2.81.

[23] B. Efron and C. Stein. The jackknife estimate of variance. *The Annals of Statistics*, 9(3):586–596, 1981. doi: 10.1214/aos/1176345462.

[24] Xiaohui Chen. Gaussian and bootstrap approximations for high-dimensional U-statistics and their applications. *The Annals of Statistics*, 46(2):642–678, 2018. doi: 10.1214/17-AOS1563. URL https://arxiv.org/abs/1610.00032.